

FAST NUCES

CS 3015 – Robotics Technology

Fall 2024 – Assignment IV

Important Dates

Date Assigned: 25th November 2024

Deadline: 29th November 2024, Friday.

Submission Procedure: Submit to your CR. CR will coordinate with me and submit all the assignments together (once) during the coordinated time.

Late submissions will not be accepted under any circumstances.

Topic	Name	Roll Number	Section	Signature
Member 1				
Member 2				
Member 3				
Member 4				

Formatting Requirements

1. Use A4 white pages with narrow margins.
2. Reports must be neat, clear, and well organized.
3. Include diagrams, figures, and examples to support your explanations.

Group Assignment Guidelines Collaboration Rules

1. This is a group assignment.
2. You are responsible for forming your groups.
3. No excuses will be accepted for incomplete contributions.
4. Each group must have Four (4) members.
5. The assignment will be evaluated as a whole; therefore, the division of work is your responsibility.
6. If your group has fewer than the required number of members, the division of work remains your responsibility.
7. Each group member is expected to contribute actively. Failure to do so may result in penalties or exclusion from the evaluation.
8. Ensure that all group members are familiar with and agree upon the division of tasks.
9. Group disputes will not be entertained. It is the group's responsibility to resolve any internal issues.
10. Group members should peer-review each other's work to maintain a consistent standard of the work's quality and presentation.

Submission Guidelines

1. Fill in the details of group members and attach this 1st page (only) as the title page.
2. The assignment must be handwritten and completed individually.
3. Group submissions should compile all individual reports into a single, organized document.
4. Plagiarized work or solutions copied without understanding will result in a penalty or no credit.
5. Ensure that the final submission includes all necessary components (reports, diagrams, computations, and references).

Assessment Criteria

1. Neatness & Presentation: 50%
2. Accuracy/Efforts of Problem Explanation: 10%
3. Correctness of Solutions: 10%
4. Use of Diagrams & Examples: 10%
5. References and Citations: 10%

Submission Checklist

Before submitting the assignment, verify the following:

- a. Is the report handwritten and no longer than four pages (for each member solution) and the total number of pages should not exceed 18?
- b. Are diagrams and examples included?
- c. Is everything clearly explained, which is understandable for a layman?
- d. Are authentic references provided?

Identification Details:

Ensure that your Roll Number, Section, Name, and Signature are written at the bottom of each page for their part. Failure to do so will result in marks deductions, and no subsequent queries about this matter will be entertained. Note: Any poorly presented or incomplete assignment will receive no credit.

Assignment Overview

This group assignment explores different important robotics concepts. Explain these concepts in the simplest form, in a way which is understandable to a layman (a common person with minimal knowledge about robotics and electronics). Provide a **concise explanation** of the problem and its **significance**. Mention the **applications, benefits and limitations/shortcomings** of these topics along with (any) small relevant **numerical example**. It can be some conversion or formula to find something or any relevant thing. Solve this small example showing step-by-step computations or procedure (if any). Each group member is assigned specific problems to work on. Each student must prepare a handwritten less than or equal to four-page report for their topics (total 4 pages for both topics, per member). The assignment emphasizes clear problem-solving steps, efficient solutions, and collaborative learning.

References:

Cite all resources used. Only authentic sources are acceptable.

List of Topics

Members and their Assigned Topics

Member 1:

- a) PWM
- b) Motors

Member 2:

- a) Encoders
- b) Gear box

Member 3:

Serial Link Communication.

- a) Protocol SPI
- b) Protocol I²C

Member 4:

- a) Interrupts
- b) Timers

--- The End ---